A Prefactor Coefficient Law for the Gauss–Legendre AGM Algorithm

A refined diagnostic asymptotic

Wayne Baker
Independent Researcher

Original: June 30, 2026; v1.2 reviewed release candidate: October 29, 2026

Abstract

For the lower Gauss–Legendre approximants π_n, consider the exact, reference-dependent diagnostic weight $\alpha_n = -(\pi_{n+1} - \pi)/(\pi_{n+1} - \pi_n)$. Using the leading two-term theta-function prefactor recorded by Brent, we recover $\alpha_{n+1}/\alpha_n^2 \to 1/2$ as a direct corollary and sharpen the limit by an explicit rational prefactor model. With $x_n = 2^{n+1}\pi$, under a quantified exponential remainder hypothesis, we obtain

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{(4x_n - 1)(x_n - 1)^2}{(2x_n - 1)^3} + O(\mathrm{poly}(2^n)e^{-x_n}) = \frac{1}{2} - \frac{3}{8x_n} + O(x_n^{-2}).$$

This explains the observed approach to 1/2 from below. A numerical investigation identifies a possible second exponential term and a correspondingly sharper finite-index correction; these are stated as observations, not theorems. The weight depends on the unknown target π and is not an implementable acceleration method.

1 AGM indexing and the diagnostic coefficient

Start with $a_0 = 1$, $b_0 = 1/\sqrt{2}$, $t_0 = 1/4$, and $p_0 = 1$, and use

$$\begin{aligned} a_{n+1} &= (a_n + b_n)/2, & b_{n+1} &= \sqrt{a_n b_n}, \\ t_{n+1} &= t_n - p_n(a_n - a_{n+1})^2, & p_{n+1} &= 2p_n. \end{aligned}$$

Throughout this paper the lower approximant is

$$\pi_n = \frac{(a_n + b_n)^2}{4t_n} = \frac{a_{n+1}^2}{t_n}, \qquad \varepsilon_n = \pi_n - \pi < 0.$$

The index on π_n is the index of a_n, b_n, t_n appearing on the left expression, not of a_{n+1}. Define

$$\alpha_n = -\frac{\pi_{n+1} - \pi}{\pi_{n+1} - \pi_n} = -\frac{\varepsilon_{n+1}}{\varepsilon_{n+1} - \varepsilon_n}.$$

Thus $\pi = \pi_{n+1} + \alpha_n(\pi_{n+1} - \pi_n)$ holds *by definition*. Since evaluation of α_n requires π, this coefficient is a retrospective diagnostic, not a computable error-correction rule or algorithmic speed claim.

2 Classical prefactor input

Brent's analysis in Section 4 ("Some Superlinearly Convergent Algorithms for π"), equation (20), and the expansion immediately following equation (21), records the leading two-term expression and a printed remainder estimate (with $q = e^{-\pi}$) [1]:

$$\pi - \pi_n = (2^{n+4}\pi^2 - 8\pi)e^{-\pi 2^{n+1}} - O\Big(2^n e^{-\pi 2^{n+2}}\Big). \tag{1}$$

Brent's lower approximant a_{n+1}^2/s_n uses $s_0 = 1/4$ and the same AGM updates as the present $\pi_n = a_{n+1}^2/t_n$ with $t_n = s_n$; therefore the discrepancy discussed here is not an indexing convention. The minus sign preceding Brent's O symbol is explicitly described there as informal sign information. We reproduce this historical statement without treating its claimed *size* as independently established: the high-precision residual analysis described in Section 4 indicates a potentially larger polynomial prefactor at the second exponential order. Resolving the discrepancy analytically remains open.

For the theorems below we need only the weaker leading asymptotic input

$$\varepsilon_n = B_n e^{-x_n}\big(1 + o(1)\big), \qquad x_n = 2^{n+1}\pi, \quad B_n = -8\pi(x_n - 1), \quad x_{n+1} = 2x_n. \tag{2}$$

More quantitatively, a remainder $O(x_n^M e^{-2x_n})$ for some fixed M is sufficient for the exponentially small error bounds stated below; its required analytic justification must not be inferred from the numerical second-order fit alone. For the rational approximation we therefore explicitly state that quantitative hypothesis. Writing $\varepsilon_n = A_n e^{-x_n}$ gives $A_n/B_n \to 1$ and $A_{n+1}/A_n \to 2$. The limiting statement follows from the leading theta-function expansion; it is not claimed as an independent AGM theorem.

3 Coefficient law and sharpened correction

Lemma 1. *If $r_n = \varepsilon_{n+1}/\varepsilon_n \to 0$, then*

$$\alpha_n = \frac{r_n}{1 - r_n}, \qquad \frac{\alpha_{n+1}}{\alpha_n^2} = \frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3}\frac{(1 - r_n)^2}{1 - r_{n+1}}.$$

Proof. Both identities follow by substitution and cancellation. □

Theorem 1 (Prefactor coefficient limit)**.** *For the above lower Gauss–Legendre sequence,*

$$\lim_{n\to\infty} \frac{\alpha_{n+1}}{\alpha_n^2} = \frac{1}{2}.$$

Proof. By (2), $r_n \to 0$. Lemma 1 reduces the ratio asymptotically to $\varepsilon_{n+2}\varepsilon_n^2/\varepsilon_{n+1}^3$. The exponential factors cancel exactly because $x_{n+2} + 2x_n - 3x_{n+1} = 0$. The remaining ratio tends to $2(1/2)^2 = 1/2$, since $A_{n+1}/A_n \to 2$. □

Corollary 1 (Finite-index rational prefactor law under quantified remainder)**.** *Assume in addition that $\varepsilon_n = B_n e^{-x_n} + O(x_n^M e^{-2x_n})$ for some fixed $M \geq 0$. Let $x = x_n = 2^{n+1}\pi$. Then*

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{(4x - 1)(x - 1)^2}{(2x - 1)^3} + O(x^K e^{-x}), \tag{3}$$

and hence

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{1}{2} - \frac{3}{8x} + O(x^{-2}) = \frac{1}{2} - \frac{3}{2^{n+4}\pi} + O(4^{-n}). \tag{4}$$

Also the rescaled error prefactors obey

$$\frac{A_{n+1}}{A_n} = 2 + \frac{1}{x-1} + O(x^K e^{-x}).$$

Proof. Use $B_n = -8\pi(x-1)$, $B_{n+1} = -8\pi(2x-1)$, and $B_{n+2} = -8\pi(4x-1)$ in Lemma 1. The pure prefactor quotient is exactly the rational expression in (3). Under the additional quantified remainder hypothesis, the relative errors in the B_k are exponentially small, and $r_n = O(e^{-x}\,\mathrm{poly}(2^n))$; their effect on the diagnostic quotient is likewise exponentially small. For a sufficiently large fixed K depending on M, expansion in $1/x$ gives $1/2 - 3/(8x) + O(x^{-2})$. The prefactor ratio follows from $B_{n+1}/B_n = (2x-1)/(x-1)$ and the same quantified hypothesis. □

Remark 1. Equation (4) describes the eventual approach to 1/2 from below. Under the stated quantitative hypothesis, the finite-n rational expression is an asymptotic model, not an exact identity for the numerical AGM coefficient. Exponentially small corrections remain.

4 Second exponential order: numerical observation and open analytic issue

Numerical Observation 1 (Numerically identified second-order coefficient)**.** Put $x = 2^{n+1}\pi$. The high-precision independent review reports that the residual after the leading prefactor is consistent, at the tested indices, with

$$\varepsilon_n = -8\pi(x-1)e^{-x} + 8\pi(2x-1)(4x-3)e^{-2x} + \text{higher exponential orders.} \tag{5}$$

The coefficient $8\pi(2x-1)(4x-3)$ was numerically identified using calculations up to 6000 decimal digits, with reported fit residual below 10^{-75}; it has *not* been derived here from the theta-function formulas. In particular (5) does not establish a uniform remainder estimate or an exact coefficient identity.

The observed second term grows as $O(x^2e^{-2x})$, whereas the printed remainder in (1) has the smaller polynomial factor $O(xe^{-2x})$. These two asymptotic descriptions cannot both be valid with the standard meaning of O if the candidate quadratic coefficient is correct. The indexing and lower-approximant definitions agree; an analytic theta-function expansion is needed to adjudicate the discrepancy. We do *not* attribute the candidate $O(x^2e^{-2x})$ remainder to Brent.

Numerical Observation 2 (Candidate exponential correction to the rational model)**.** Propagation of the identified second-order term through Lemma 1 suggests

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{(4x-1)(x-1)^2}{(2x-1)^3} - \frac{4(4x-1)(x-1)}{2x-1}e^{-x} + \text{smaller terms.} \tag{6}$$

The independent numerical comparison reports residuals of approximately 1.7×10^{-7}, 5.2×10^{-39} and 8.1×10^{-343} for $n = 1, 3, 6$, respectively, after this correction. The further claim that the omitted term is $O(x^2e^{-2x})$ remains *conditional* on a rigorously controlled third-order theta expansion; it is not asserted as a theorem in this revision.

5 Numerical verification

The following figures are retained from the original manuscript; they were reported with 500-decimal-digit `mpmath` calculations and the above indexing. The theorem is retained; the additional second-order statements are explicitly numerical and do not replace that earlier numerical record.

n	α_n
0	$4.476840443051657 \times 10^{-3}$
1	$7.278747106485182 \times 10^{-6}$
2	$2.482705775153228 \times 10^{-11}$
3	$2.988090953871379 \times 10^{-22}$
4	$4.397065311352923 \times 10^{-44}$
5	$9.594611535813435 \times 10^{-88}$
6	$4.585616315217315 \times 10^{-175}$

Table 1: Reference-dependent fitted diagnostic weights.

n	α_{n+1}/α_n^2	A_{n+1}/A_n
0	0.3631728700311416	2.3866261558728965
1	0.4686104763344650	2.0871750987545307
2	0.4847784478028806	2.0414374856886031
3	0.4924649101466708	2.0202981874959604
4	0.4962511920458477	2.0100471245391349
5	0.4981302572056423	2.0049984522335209

Table 2: Observed coefficient and prefactor ratios.

6 Numerator channel and computational scope

Set $q_n = (a_n + b_n)^2$ and

$$\beta_n = -\frac{q_{n+1} - 4\pi t_{n+1}}{q_{n+1} - q_n}.$$

Numerical Observation 3 (Numerical only)**.** The original high-precision calculation suggested $\beta_{n+1}/\beta_n^2 \to 0$. An independent high-precision review reproduced the reported values and found them consistent with $\beta_{n+1}/\beta_n^2 \sim -1/x_n$. Neither this asymptotic equivalent nor the limit is proved here.

The static Baker N-series uses fixed rational weights to remove powers of a resolution parameter. In contrast the present α_n uses an unknown limit and is *not* an acceleration prescription. Aitken's Δ^2 extrapolation is a different, implementable three-point operation; its merit relative to taking the next AGM iterate is not asserted here without a separate error-and-cost comparison.

7 Conclusion

The limit $\alpha_{n+1}/\alpha_n^2 \to 1/2$ is a transparent consequence of Brent's lower-approximant asymptotic. Retaining the subleading -8π prefactor term motivates the rational model (3); its stated exponential error control is conditional on a quantified lower-approximant remainder. A new numerical candidate for the second exponential term exposes an unresolved discrepancy with the polynomial factor in a printed remainder estimate and motivates a future theta-function derivation. This is a diagnostic comparison of error structure, not a new algorithm for π or a fixed-coefficient extrapolation method.

References

[1] R. P. Brent, "The Borwein brothers, Pi and the AGM," in *From Analysis to Visualization*, Springer Proceedings in Mathematics & Statistics 313 (2020), 323–348; also arXiv:1802.07558.

[2] R. P. Brent, "Fast multiple-precision evaluation of elementary functions," *J. ACM* 23 (1976), 242–251.

[3] J. M. Borwein and P. B. Borwein, *Pi and the AGM*, Wiley, 1987.

[4] E. Salamin, "Computation of π using arithmetic-geometric mean," *Math. Comp.* 30 (1976), 565–570.

[5] A. C. Aitken, "On Bernoulli's numerical solution of algebraic equations," *Proc. Roy. Soc. Edinburgh* 46 (1926), 289–305.